

COMPUTER NETWORKS LAB

DIGITAL ASSIGNMENT-2

SUBMITTED BY:
APOORVA KARNWAL 22BCE0756

1.HAMMING CODE

CODE:

```
// APOORVA KARNWAL
// 22BCE0756
#include <stdio.h>
#include <math.h>
int input[32];
int code[32];
int ham_calc(int, int);
void main(){
    int n, i, p_n = 0, c_l, j, k;
    printf("Please enter the length of the Data Word: ");
    scanf("%d", &n);
    printf("Please enter the Data Word:\n");
    for (i = 0; i < n; i++){
        scanf("%d", &input[i]);
    }
    i = 0;
    while (n > (int)pow(2, i) - (i + 1)){
        p_n++;
        i++;
    }
    c_l = p_n + n;
    j = k = 0;
    for (i = 0; i < c_l; i++){
        if (i == ((int)pow(2, k) - 1)){
            code[i] = 0;
            k++;
        }
        else{
            code[i] = input[j];
            j++;
        }
    }
    for (i = 0; i < p_n; i++){
        int position = (int)pow(2, i);
        int value = ham_calc(position, c_l);
        code[position - 1] = value;
    }
}
```

```

printf("\nThe Calculated Code Word: ");
for (i = 0; i < c_; i++)
    printf("%d", code[i]);
printf("\n");
printf("Enter the received Codeword:\n");
for (i = 0; i < c_; i++)
    scanf("%d", &code[i]);
int pos = 0;
for (i = 0; i < p_n; i++){
    int position = (int)pow(2, i);
    int value = ham_calc(position, c_);
    if (value != 0)
        pos += position;
}
if (pos == 1)
    printf("The received Code Word is correct.\n");
else
    printf("Error at bit position: %d\n", pos);
}
int ham_calc(int position, int c_){
    int count = 0, i, j;
    i = position - 1;
    while (i < c_){
        for (j = i; j < i + position; j++){
            if (code[j] == 1)
                count++;
        }
        i = i + 2 * position;
    }
    if(count%2 == 0)
        return 0;
    else
        return 1;
}

```

OUTPUT:

```

Code - L2_patterns
Please enter the length of the Data Word: 3
Please enter the Data Word:
1
0
1

The Calculated Code Word: 101101
Enter the received Codeword:
1
1
1
1
0
1

Error at bit position: 2
(base) karnwalpoorva23@Apoorvas-MacBook-Air L2_patterns %

```

CRC

CODE:

```
// APOORVA KARNWAL
// 22BCE0756
#include <stdio.h>
#include <string.h>
#define MAX_LENGTH 50
char number[MAX_LENGTH];
char check[MAX_LENGTH];
char code_word[MAX_LENGTH];
int number_length, i, j;

void XOR(int n) {
    for (j = 0; j < n; j++) {
        check[j] = (check[j] == code_word[j]) ? '0' : '1';
    }
}

void crc(int n) {
    for (i = 0; i < n; i++) {
        check[i] = number[i];
    }

    for (i = n; i < number_length; i++) {
        if (check[0] == '1') {
            XOR(n);
        }
        for (j = 0; j < n - 1; j++) {
            check[j] = check[j + 1];
        }
        check[j] = number[i];
    }
    if (check[0] == '1') {
        XOR(n);
    }
}

int main() {

    int n;
    printf("\nEnter data to be transmitted: ");
    scanf("%s", number);
    printf("\nEnter the code word: ");
    scanf("%s", code_word);
    n = strlen(code_word);
    number_length = strlen(number);

    for (i = number_length; i < number_length + n - 1; i++) {
        number[i] = '0';
    }
}
```

```

number[i] = '\0';
printf("\nAugmented number: %s\n", number);
crc(n);
printf("\nRemainder is : %s\n", check);

for (i = number_length, j = 0; j < n - 1; i++, j++) {
    number[i] = check[j];
}

number[i] = '\0';
printf("\nData to be sent : %s\n", number);
char received_number[MAX_LENGTH];
printf("Enter the received data: ");
scanf("%s", received_number);
printf("\nData received: %s\n", received_number);
strcpy(number, received_number);
number_length = strlen(number);

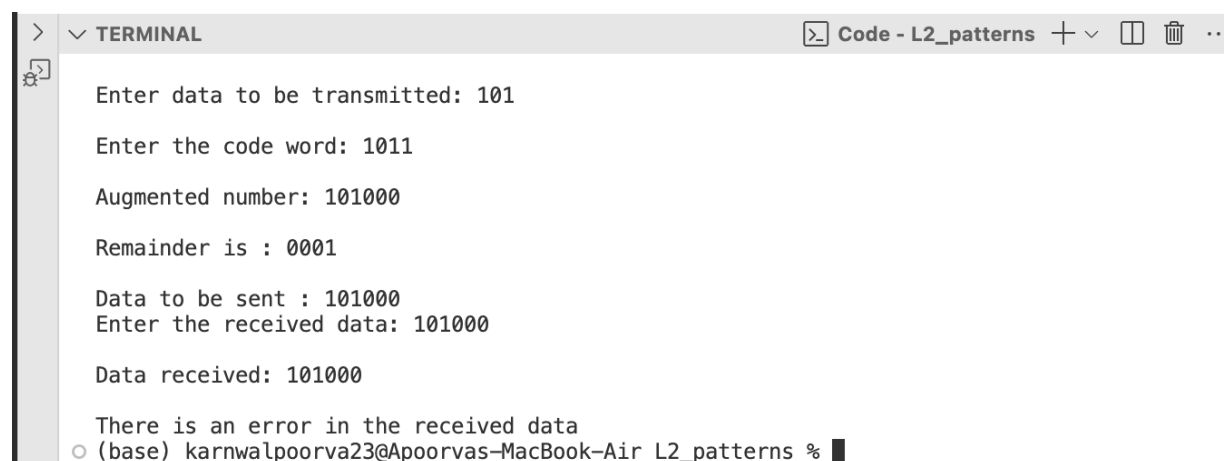
crc(n);

for (i = 0; (i < n - 1) && (check[i] != '1'); i++) {
}

if (i < n - 1) {
    printf("\nThere is an error in the received data\n");
} else {
    printf("\nReceived data matches with sender data\n");
}
return 0;
}

```

OUTPUT:



A terminal window titled "Code - L2_patterns" showing the execution of a C program. The user enters "101" for data and "1011" for the code word. The program outputs the augmented number "101000" and the remainder "0001". Then, the user enters "101000" for the received data, and the program outputs "Data received: 101000". Finally, the program outputs "There is an error in the received data" because the received data does not match the augmented number.

```

> ▾ TERMINAL Code - L2_patterns + ▾ 🗑️ ⋮
Enter data to be transmitted: 101
Enter the code word: 1011
Augmented number: 101000
Remainder is : 0001
Data to be sent : 101000
Enter the received data: 101000
Data received: 101000
There is an error in the received data
(base) karnwalpoorva23@Apoorvas-MacBook-Air L2_patterns %

```

CHECKSUM

CODE:

```
// APOORVA KARNWAL
// 22BCE0756
#include <stdio.h>
#include <math.h>
int sender(int arr[10], int n){
    int checksum, sum = 0, i;
    printf("\n****SENDER SIDE****\n");
    for (i = 0; i < n; i++)
        sum += arr[i];
    printf("SUM IS: %d", sum);
    checksum = ~sum; // 1's complement of sum
    printf("\nCHECKSUM IS:%d",checksum); return checksum;
}
void receiver(int arr[10], int n, int sch){
    int checksum, sum = 0, i;
    printf("\n\n****RECEIVER SIDE****\n");
    for (i = 0; i < n; i++)
        sum += arr[i];
    printf("SUM IS:%d", sum);
    sum = sum + sch;
    checksum = ~sum; // 1's complement of sum
    printf("\nCHECKSUM IS:%d",checksum);
}
void main(){
    int n, sch, rch;
    printf("\nENTER SIZE OF THE STRING:");
    scanf("%d", &n);
    int arr[n];
    printf("ENTER THE ELEMENTS OF THE ARRAY TO CALCULATE CHECKSUM:\n");
    for (int i = 0; i < n; i++){
        scanf("%d", &arr[i]);
    }
    sch = sender(arr, n);
    receiver(arr, n, sch);
}
```

OUTPUT:

```

v TERMINAL Code - L2_patterns + v [] [X] ..
ENTER SIZE OF THE STRING:3
ENTER THE ELEMENTS OF THE ARRAY TO CALCULATE CHECKSUM:
13
23
11

****SENDER SIDE****
SUM IS: 47
CHECKSUM IS:-48

****RECEIVER SIDE****
SUM IS:47
CHECKSUM IS:0%
o (base) karnwalpoorva23@Apoorvas-MacBook-Air L2_patterns %
```